Scientific inference and evidence graphs

From a paper collection to testable new research

This original Atlas teaching brief introduces scientific foundations and links to primary literature. It is not a third-party research paper, peer-reviewed original result, or replacement for source documents.

What counts as emergent insight?

Knowledge discovery and literature-based discovery can reveal connections between ideas. A graph-completed edge or AI-generated analogy is a proposal until it survives evaluation.

Preserve exact scientific context

Carry source/version ID, equation, defined variables, units, assumptions, boundary conditions, measurement process, uncertainty, and the strongest competing explanation.

Hypothesis and controls

Create a falsifiable prediction, a simple null model and negative controls. Test on synthetic or public authorized data. Report failures alongside successes.

Status gates

Candidate -> reproduced -> independently corroborated -> accepted with scope. Each transition needs an evidence trail and review. Similar terms in two papers do not justify a new law of nature.

Possible study

Two papers discuss field-based models and information theory. Extract their mathematical assumptions and test whether the same simplified channel equation fits both data models; document discrepancies.

Sources and further reading

PUB-013: W3C PROV-O (2013) establishes vocabulary for provenance. PUB-001 and PUB-005 provide scientific starting points for the cross-domain example; it is not a confirmed breakthrough.